

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA1710

COURSE NAME: Artificial Intelligence

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
Total Marks: 30

QUESTIONS: 5

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I G. Pallavi (192311208) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pallavi

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Student 6: GUDE PALLAVI | Register Number: 192311208

Question 1 - Flight Delay Notification

Knowledge Base

1. If a flight is delayed, then passengers are notified.
2. Flight 202 is delayed.

Goal

Prove that passengers are notified, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 2 - Employee Certification

Knowledge Base

1. If an employee completes the training, then the employee receives a certification.
2. Kevin completed the training.

Goal

Prove that Kevin receives a certification, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 3 - Medical Attention Alert

Knowledge Base

1. If a patient has a fever, then the patient needs medical attention.
2. Ravi has a fever.

Goal

Prove that Ravi needs medical attention, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 4 - Library Late Fine

Knowledge Base

1. If a book is returned late, then a fine is charged.
2. The book was returned late.

Goal

Prove that a fine is charged, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 5 - Vehicle Registration Process

Knowledge Base

1. If a car passes the emissions test, then the car is certified roadworthy.
2. If a car is certified roadworthy, then the car can be registered.
3. The car passed the emissions test.

Goal

Prove that the car can be registered. using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (Box).
5. Explain each resolution step and write the final conclusion.

Flight Delay Notification :-

1) Propositional Logic :-

Let

$\rightarrow D$ = Flight 202 is delayed.

$\rightarrow N$ = Passengers are notified.

Knowledge base :

1. $D \rightarrow N$

2. D

Goal : N

2) CNF and Negated Goal :-

$\rightarrow D \rightarrow N = \neg D \vee N$

$\rightarrow D = D$

$\rightarrow$ Negated Goal = $\neg N$

3) Resolution Steps :-

Step 1 :-

$(\neg D \vee N)$ and $D \Rightarrow N$

Step 2 :-

N and $\neg N$

$\Rightarrow$ $\square$ Empty Clause

Conclusion :-

Since the Empty clause is obtained, Passengers are notified is proved.

2. Employee Certification:-

1. Propositional Logic :

Let

• T = Kevin completed the training

• C = Kevin receives certification

Knowledge base :

1. $T \rightarrow C$

2. T

Goal : C

2. CNF and Negated Goal :-

• $T \rightarrow C = \neg T \vee C$

• $T = T$

• Negated goal = $\neg C$

3. Resolution Steps :

Step 1:- $(\neg T \vee C)$ and $T \Rightarrow C$

Step 2:- C and $\neg C$

$\Rightarrow \neg$ Empty clause

Conclusion :-

Therefore, Kevin receives certification is proved.

3. Medical Attention Alert :-

1) Propositional Logic :-

Let :

$\rightarrow F$ = Ravi has a fever.

$\rightarrow M$ = Ravi needs medical attention.

Base :
1. $F \rightarrow M$
2. F

Goal : M

2. CNF and Negated Goal :-

• $F \rightarrow M = \neg F \vee M$

• $F = F$

• Negated goal = $\neg M$

3. Resolution Steps:-

Step 1:-

$(\neg F \vee M)$ and F

$\Rightarrow M$

Step 2:-

M and $\neg M$

$\Rightarrow \square$ Empty clause

Conclusion :- $\therefore$ Ravi needs medical attention is proved.

4. Library Late Fine :-

(1) Propositional Logic :-

Let :

• L = Book is returned late

• F = Fine is charged.

Knowledge Base :

1. $L \rightarrow F$

2. L

Goal :-

2) CNF and Negated Goal :-

• $L \rightarrow F \rightarrow \neg L \vee F$

• $L \vee L$

• Negated goal :- $\neg F$

3) Resolution Steps :-

Step 1:- $(\neg L \vee F)$ and L $\Rightarrow F$

Step 2 :- F and $\neg F$

$\Rightarrow \square$ Empty clause

Conclusion :- Therefore, a fine is charged is proved.

6. Vehicle Registration Process:-

(1) Propositional logic :-

Let :

• E = Car passed the emissions test.

• R = Car is certified roadworthy.

• G = Car can be registered.

Knowledge base :

1. $E \rightarrow R$

2. $R \rightarrow G$

3. E

Goal = G

Convert into CNF :-

Step 1 :-

$$E \rightarrow R$$

CNF :

$$\neg E \vee R$$

Rule 2 :-

$$R \rightarrow G$$

CNF :

$$\neg R \vee G$$

Fact : E

(3) Negate the Goal :-

Goal : G

Negated Goal : $\neg G$

(4) Resolution Steps :-

Step 1 :-

Resolve : $(\neg E \vee R)$ and E

Result : R

The Car is certified trustworthy.

Step 2 :-

Resolve :-

$(\neg R \vee G)$ and R

Result :- G

RUBRICS FOR CALCULATION

Numerical Problem					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding, misses key elements	Minimal understanding or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps, multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors, incorrect final answer	No calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation